BIO 004 Human Anatomy · Module 4 notes · Dr. Sharilyn Rennie

The meninges and cerebrospinal fluid

The written notes for the meninges and cerebrospinal fluid lecture: the cranial and spinal meninges, the dural reflections and venous sinuses, the meningeal spaces, the end of the cord and the lumbar puncture, the ventricles, the CSF route, and the blood-brain barrier. Module 5 competencies 167 to 172, with eight worked questions.

To print: these notes as a two-column PDF (opens in a new tab). Already printed the Module 4 notes packet? These are the lecture notes, a different document, so print these too if you want them on paper.

How to use these notes. Draw the note sheet for each competency first, from memory, before you read. Then read the matching section here to fill in anything you missed, and use it to review. Every section carries its competency number, so you can go from the competency packet to the page and back. The worked questions at the end show how a lecture exam asks about this material and why each wrong answer is wrong.

Competency 167. Cranial meninges and dural reflections

In your course notes: Module 5 packet, Chapter 6, competency 167 (page 48).

Dr. Rennie's video for this part: Meninges and CSF concept video.

You should be able to: Identify the dura, arachnoid, and pia mater and the dural reflections falx cerebri, falx cerebelli, and tentorium cerebelli, and trace dural venous sinus drainage from the superior sagittal sinus to the internal jugular vein.

Three connective tissue membranes, the meninges, wrap the brain. From outside to inside they are the dura mater, the arachnoid mater, and the pia mater. The names describe them. Dura mater means tough mother, pia mater means tender mother, and arachnoid means spider-like, for the web of fine strands it sends down to the pia.

The three layers, from the skull to the brain

Dura mater

The outermost layer, against the skull. Around the brain the dura has two layers. The periosteal layer is attached to the inner surface of the skull. The meningeal layer lies inside it. The two layers are fused together except in a few places where they separate, and the spaces between them form the dural venous sinuses, channels that collect venous blood from the brain.

Arachnoid mater

The thin, delicate middle membrane just inside the dura. It does not dip into the grooves of the brain. Fine web-like strands cross from it to the pia, through the CSF-filled space between them.

Pia mater

The innermost layer, a very thin membrane rich in small blood vessels. It clings to the surface of the brain and follows it into every sulcus.

Layer	Position	Description
Dura mater	Outermost, against the skull	Tough; two layers (periosteal and meningeal) fused except where they part to form the dural venous sinuses
Arachnoid mater	Middle	Thin and delicate; sends web-like strands to the pia; does not follow the sulci
Pia mater	Innermost, on the brain surface	Very thin and rich in blood vessels; follows every gyrus and sulcus

- On a strip drawn from skull to brain the order is dura, arachnoid, pia, with the pia clinging to the brain surface.
- The dura is drawn as two layers: the periosteal layer against the skull and the meningeal layer beneath it. The arachnoid and pia are each a single layer.

The dural reflections

In three main places the meningeal layer of the dura folds inward as a sheet that divides the cranial cavity and steadies the brain. These folds are the dural reflections, also called the dural extensions. The falx cerebri drops into the longitudinal fissure between the right and left cerebral hemispheres. The tentorium cerebelli lies in the transverse fissure, like a tent over the cerebellum, separating it from the occipital lobes of the cerebrum above. The falx cerebelli is a small fold below the tentorium that separates the right and left cerebellar hemispheres.

Fold	Where it is	What it separates
Falx cerebri	In the longitudinal fissure, along the midline	Right and left cerebral hemispheres
Tentorium cerebelli	In the transverse fissure, roofing the cerebellum	Cerebrum above from cerebellum below
Falx cerebelli	Midline, below the tentorium	Right and left cerebellar hemispheres

- The tentorium cerebelli separates the cerebrum from the cerebellum. It does not split the cerebellum. Splitting the cerebellum into right and left halves is the job of the falx cerebelli.
- Falx means sickle. Both falx folds are midline sheets that separate a right half from a left half; one is for the cerebrum and the other, much smaller, is for the cerebellum.

Dural venous sinus drainage

The main sinuses run in the edges of these folds. The superior sagittal sinus runs along the top edge of the falx cerebri, where it attaches to the skull. The inferior sagittal sinus runs in the lower free edge of the falx cerebri. It drains into the straight sinus, which runs where the falx cerebri meets the tentorium cerebelli. The superior sagittal and straight sinuses meet at the back of the skull, at the confluence of sinuses. Follow the blood out of the skull from the superior sagittal sinus.

1. Superior sagittal sinus: runs from front to back along the top edge of the falx cerebri. The arachnoid granulations return CSF into it.
2. Confluence of sinuses: the meeting point at the back of the skull, at the internal occipital protuberance. The straight sinus also drains here.
3. Transverse sinuses: paired sinuses that run laterally from the confluence along the back edge of the tentorium cerebelli.
4. Sigmoid sinuses: S-shaped sinuses that curve down from each transverse sinus to the jugular foramen.
5. Internal jugular veins: the sigmoid sinus passes through the jugular foramen and becomes the internal jugular vein, which carries the blood down the neck toward the heart.

- Blood runs front to back along the superior sagittal sinus, then out to the side in the transverse sinus, then down the sigmoid sinus and into the internal jugular vein.
- The arachnoid granulations empty CSF into the superior sagittal sinus, not into the internal jugular vein. By the time the blood reaches the jugular vein the CSF is already mixed into it.

Competency 168. Spinal meninges and meningeal spaces

In your course notes: Module 5 packet, Chapter 6, competency 168 (page 49).

Dr. Rennie's video for this part: Meninges and CSF concept video.

You should be able to: Identify the spinal meninges, the denticulate ligaments and filum terminale, and distinguish the epidural, subdural, and subarachnoid spaces by location and contents.

The same three meninges continue down the vertebral canal around the spinal cord. The spinal dura mater is a single layer. It matches the meningeal layer of the cranial dura and is not attached to the vertebrae. It forms a tube, the dural sac, that ends at about the level of S2. The spinal arachnoid mater lines the dural sac. The spinal pia mater clings to the surface of the cord.

Two anchors made of pia

Denticulate ligaments

Tooth-shaped flaps of pia along each side of the cord, between the dorsal and ventral roots. They reach out through the arachnoid and attach to the dura, which keeps the cord centered in the dural sac. They anchor the cord to the dura along its length.

Filum terminale

A thin strand of pia that continues down from the bottom tip of the cord. It passes through the lumbar cistern, picks up a covering of dura at the end of the dural sac, and anchors to the coccyx. It anchors the lower end of the cord.

- Both anchors are extensions of the pia mater. The denticulate ligaments tie the sides of the cord to the dura; the filum terminale ties the end of the cord to the coccyx.

The meningeal spaces

The epidural space lies between the dura and the bone. In the vertebral canal it is a real space, filled with fat and veins, and it is where an epidural injection goes. In the skull the periosteal dura is attached to the bone, so the cranial epidural space is only a potential space. It opens only if something, such as blood, separates the dura from the skull.

The subdural space lies between the dura and the arachnoid. Normally the two membranes touch, so it is a potential space, holding only a thin film of fluid. The subarachnoid space lies between the arachnoid and the pia. It is a real space. It holds cerebrospinal fluid, the web-like strands of the arachnoid, and the larger blood vessels that run over the surface of the brain and cord.

Space	Location	Contents	Found in the cranium too?
Epidural space	Between the dura and the bone	Spinal: fat and veins (a real space). Cranial: a potential space only	Only as a potential space, because the dura is attached to the skull
Subdural space	Between the dura and the arachnoid	A potential space with a thin film of fluid	Yes
Subarachnoid space	Between the arachnoid and the pia	Cerebrospinal fluid, arachnoid strands, and the larger surface blood vessels	Yes

- On a cross section of a vertebra, the order from the outside in is vertebral bone, epidural space (fat and veins), dura mater, subdural space, arachnoid mater, subarachnoid space (CSF), pia mater on the surface of the cord.
- CSF is in the subarachnoid space, under the arachnoid, not in the epidural or subdural space.
- The epidural space is a real space only in the spine. In the cranium the dura is attached to the skull, so there is no room there unless something forces the two apart.

Disorders named by meningeal layer or space

Bleeding and infection are named by the layer or space they involve.

Disorder	Structure affected	What changes there
Epidural hematoma	Between the skull and the dura	Blood, often from an artery torn by a skull fracture, separates the dura from the bone
Subdural hematoma	Subdural space	Blood, often from torn veins crossing to the dural sinuses, collects between the dura and the arachnoid
Subarachnoid hemorrhage	Subarachnoid space	Blood from a ruptured surface artery mixes into the cerebrospinal fluid
Meningitis	The meninges, mainly the arachnoid and pia	The membranes become inflamed, usually from an infection; the CSF in the subarachnoid space shows the change

What this means for patients

- An epidural hematoma opens the cranial epidural space, a space that normally does not exist. The blood has to peel the dura off the skull, which is why it is usually arterial blood under pressure.
- A subdural hematoma sits just beneath the dura, in the potential space between the dura and the arachnoid, usually from veins torn where they cross to the dural sinuses.
- A subarachnoid hemorrhage and meningitis both change the CSF, because both involve the subarachnoid space. That is why CSF can be tested for either one.
- An epidural injection goes into the spinal epidural space, the fat-filled and vein-filled space outside the dura. It never enters the CSF.

Competency 169. Cord termination and lumbar puncture

In your course notes: Module 5 packet, Chapter 6, competency 169 (page 50).

Dr. Rennie's video for this part: Meninges and CSF concept video.

You should be able to: Identify the conus medullaris, cauda equina, and lumbar cistern and list in order the layers a lumbar puncture needle crosses from skin to subarachnoid space.

The spinal cord is shorter than the vertebral column. In an adult it ends at about the level of the disc between L1 and L2. The tapered, cone-shaped end of the cord is the conus medullaris.

The nerve roots for the lumbar, sacral, and coccygeal spinal nerves leave the cord above that level. They have to travel down the vertebral canal before they reach their exit openings. This bundle of long roots below the conus medullaris is the cauda equina, which means horse's tail.

The dural sac and the subarachnoid space continue down to about S2. Below the conus medullaris the subarachnoid space is wide and holds no spinal cord, only CSF, the roots of the cauda equina, and the filum terminale. This enlarged space is the lumbar cistern.

Structure	Level	What it is
Conus medullaris	L1 to L2 disc (adult)	The tapered lower end of the spinal cord
Cauda equina	Below L1 to L2, down the vertebral canal	The bundle of lumbar, sacral, and coccygeal nerve roots
Filum terminale	From the conus medullaris to the coccyx	A strand of pia mater that anchors the lower end of the cord
Lumbar cistern	From the conus medullaris to about S2	The enlarged subarachnoid space, filled with CSF and the cauda equina

- The cord ends at the L1 to L2 disc, not at the bottom of the spine. On a drawing, the cord stops at the conus and only roots, the filum terminale, and CSF continue below it.

The lumbar puncture

A lumbar puncture, or spinal tap, draws CSF from the lumbar cistern. The needle goes in at the midline between the spinous processes of L3 and L4 or L4 and L5, well below the end of the cord. A line drawn across the tops of the iliac crests crosses the spine at about L4, which helps find the level. The roots of the cauda equina float in the CSF and slide aside from the needle. These are the layers the needle crosses, superficial to deep.

1. Skin: the epidermis and dermis of the lower back.
2. Subcutaneous tissue: the fatty layer under the skin.
3. Supraspinous ligament: runs along the tips of the spinous processes.
4. Interspinous ligament: fills the gap between neighboring spinous processes.
5. Ligamentum flavum: the tough, elastic ligament joining the laminae of neighboring vertebrae. The needle meets firm resistance here and then a give.
6. Epidural space: the fat-filled and vein-filled space between the bone and the dura.
7. Dura mater: the tough outer meningeal membrane.
8. Arachnoid mater: the thin middle membrane, pressed against the dura, so the needle passes the two together.
9. Subarachnoid space: the needle tip is now in the lumbar cistern, and CSF can be drawn.

- There are three ligaments in a row, and the ligamentum flavum is the last and deepest of them. It sits just above the epidural space.
- The firm ligamentum flavum giving way is the landmark the clinician feels. After that come the epidural space, the dura, and the arachnoid, and then CSF appears.
- The entry levels are between the L3 and L4 spines or the L4 and L5 spines. The line across the tops of the iliac crests marks about L4.

What this means for patients

- The cord ends, but the fluid does not. The spinal cord stops at the conus medullaris near the L1 to L2 disc, yet the subarachnoid space runs lower, as the lumbar cistern. This difference in length is the basis of the lumbar puncture. A needle placed between the L3 and L4 or L4 and L5 spines reaches a pool of CSF where only the floating cauda equina lies in its path, and those roots drift aside.
- Knowing the layers in order is what makes the tap safe. The needle crosses skin, subcutaneous tissue, three ligaments, the epidural space, and then the dura and arachnoid before CSF appears.
- The CSF drawn this way is tested for signs of meningitis or bleeding.
- The named spaces also tell bleeds apart: an epidural hematoma lies between the skull and the dura, a subdural hematoma lies just beneath the dura, and a subarachnoid hemorrhage mixes into the CSF itself.

Competency 170. Ventricular system

In your course notes: Module 5 packet, Chapter 6, competency 170 (page 51).

Dr. Rennie's video for this part: Meninges and CSF concept video.

You should be able to: Identify the lateral, third, and fourth ventricles and the central canal and describe how the interventricular foramina, cerebral aqueduct, and fourth ventricle apertures connect them.

The ventricles are connected, fluid-filled spaces inside the brain. There are four: two lateral ventricles, the third ventricle, and the fourth ventricle. They are lined by ependymal cells, a type of glial cell, and filled with cerebrospinal fluid. Below the fourth ventricle the same system continues as the central canal of the spinal cord.

The four ventricles and the central canal

Lateral ventricles

The largest. There is one in each cerebral hemisphere. Each is C-shaped, curving from the frontal lobe back and then down into the temporal lobe. In the midline the two lateral ventricles are separated by a thin sheet called the septum pellucidum.

Third ventricle

A narrow slit in the midline of the diencephalon, between the right and left halves of the thalamus. Each lateral ventricle opens into it through a small hole, the interventricular foramen (foramen of Monro). There is one foramen on each side.

Cerebral aqueduct

Also called the aqueduct of Sylvius. The narrow channel through the midbrain. The third ventricle drains down through it into the fourth ventricle.

Fourth ventricle

Lies behind the pons and upper medulla and in front of the cerebellum, so it sits between the brainstem and the cerebellum.

Central canal

The narrow channel that runs down the center of the spinal cord for its full length. It is a direct continuation of the bottom of the fourth ventricle.

How the fourth ventricle empties

The fourth ventricle has two kinds of exits. Three apertures in its roof and sides open into the subarachnoid space around the brain: one median aperture (foramen of Magendie) in the midline and two lateral apertures (foramina of Luschka), one on each side. Separately, the bottom of the fourth ventricle narrows directly into the central canal, which runs down the center of the spinal cord. The apertures lead only to the subarachnoid space, and the central canal opens only from the fourth ventricle itself.

Space	Location	Connects to	Through
Lateral ventricles	One in each cerebral hemisphere; separated by the septum pellucidum	Third ventricle	Interventricular foramina (of Monro)
Third ventricle	Midline of the diencephalon, between the halves of the thalamus	Fourth ventricle	Cerebral aqueduct (of Sylvius)
Fourth ventricle	Behind the pons and upper medulla, in front of the cerebellum	Subarachnoid space	One median aperture and two lateral apertures
Fourth ventricle	Lower end, in the medulla	Central canal	Continues directly down into the cord
Central canal	Center of the spinal cord, full length	Fourth ventricle above	Direct continuation

- Memory tip: count the openings out of the fourth ventricle into the subarachnoid space: one median and two lateral, three in all. The central canal is a straight continuation down into the cord.
- The cerebral aqueduct joins the third ventricle to the fourth. The lateral ventricles reach the third ventricle through the interventricular foramina instead.
- The apertures open into the subarachnoid space, not into the central canal. The central canal joins the fourth ventricle directly, with no aperture.
- Region by region: the lateral ventricles sit in the cerebrum, the third ventricle in the diencephalon, the aqueduct in the midbrain, and the fourth ventricle between the brainstem and the cerebellum. Ependymal cells line every chamber.
- The three narrow connection points are the interventricular foramina, the cerebral aqueduct, and the apertures of the fourth ventricle. If one is blocked, CSF backs up behind it, the ventricles swell, and pressure in the skull rises. That condition is hydrocephalus, covered in competency 171.

Competency 171. CSF production and circulation

In your course notes: Module 5 packet, Chapter 6, competency 171 (page 52).

Dr. Rennie's video for this part: Meninges and CSF concept video.

You should be able to: Trace CSF from production at the choroid plexus through the ventricles and subarachnoid space to reabsorption at the arachnoid granulations.

Cerebrospinal fluid (CSF) is a clear, watery fluid that fills the ventricles and central canal and surrounds the brain and spinal cord in the subarachnoid space. It floats the brain, which lowers the brain's effective weight. It cushions the brain and cord against jolts. It also keeps a stable chemical environment around the neurons.

Where CSF is made and where it leaves

CSF is made by the choroid plexus. A choroid plexus is a tuft of capillaries covered by ependymal cells that hangs

into a ventricle. The ependymal cells make CSF from the blood plasma in the capillaries. There is choroid plexus in all four ventricles, and the largest amounts are in the lateral ventricles. About 150 mL of CSF surrounds the CNS at any one time, and about 500 mL is made each day, so the fluid is replaced several times a day.

CSF moves one way through the ventricles, out into the subarachnoid space, and back into the blood. It is reabsorbed at the arachnoid granulations, small knobs of arachnoid mater that push through the dura into the dural venous sinuses, mainly the superior sagittal sinus. From there the fluid leaves the skull with the venous blood.

The route, in order

1. Choroid plexus: capillaries covered by ependymal cells make CSF in each ventricle, most of it in the lateral ventricles.
2. Lateral ventricles: CSF collects in the ventricle in each cerebral hemisphere.
3. Interventricular foramina: CSF passes through these paired openings (foramina of Monro) into the third ventricle.
4. Third ventricle: in the midline of the diencephalon; its choroid plexus adds more CSF.
5. Cerebral aqueduct: CSF flows down this narrow channel through the midbrain.
6. Fourth ventricle: behind the pons and upper medulla; its choroid plexus adds more CSF. A small amount continues straight down into the central canal of the spinal cord.
7. Median and lateral apertures: CSF leaves the fourth ventricle through one median aperture and two lateral apertures.
8. Subarachnoid space: CSF bathes the outside of the brain and spinal cord, including the lumbar cistern.
9. Arachnoid granulations: CSF passes through these knobs of arachnoid mater into the venous blood.
10. Dural venous sinuses: the CSF now travels with the venous blood, mainly in the superior sagittal sinus, and drains out of the skull through the internal jugular veins.

- Only the three kinds of ventricle (lateral, third, and fourth) hold choroid plexus. The foramina, the aqueduct, and the apertures are passageways with no choroid plexus.
- The side branch into the central canal leaves from the fourth ventricle, not from the subarachnoid space.
- The ependymal cells of the choroid plexus are the cells that make the CSF. The capillaries supply the plasma they make it from.

Feature	Description
Where CSF is found	The four ventricles, the central canal, and the subarachnoid space around the brain and cord, including the lumbar cistern
Site of production	The choroid plexus in the lateral, third, and fourth ventricles
Site of reabsorption	The arachnoid granulations, into the dural venous sinuses
Buoyancy	Floats the brain so it does not press down on its own base
Protection	Cushions the brain and spinal cord against jolts and blows
Chemical stability	Carries nutrients, removes wastes, and keeps a steady environment around the neurons

A disorder of CSF circulation

The narrow points on this route are the interventricular foramina, the cerebral aqueduct, and the apertures of the fourth ventricle. If CSF is blocked at one of them, or is not reabsorbed, it builds up. This condition is hydrocephalus.

Disorder	Structure affected	What changes there
Hydrocephalus	The ventricles	CSF builds up behind a blockage or from poor reabsorption; the ventricles enlarge and pressure inside the skull often rises

- Memory tip: CSF is made inside the brain, in the ventricles, and reabsorbed outside the brain, at the arachnoid granulations in the dural sinuses. Anything that blocks a narrow point between those two places backs the fluid up.

What this means for patients

- Because about 500 mL is made each day and only about 150 mL is present at any moment, a blockage shows up quickly. The ventricles upstream of the block swell and the pressure inside the skull rises.
- Hydrocephalus can come from a block at any of the three narrow points, or from the arachnoid granulations failing to return CSF to the blood. Either way the fluid has nowhere to go.

Competency 172. Blood-brain barrier structure

In your course notes: Module 5 packet, Chapter 6, competency 172 (page 54).

Dr. Rennie's video for this part: Meninges and CSF concept video.

You should be able to: Name the three structural components of the blood-brain barrier: the tight-junctioned capillary endothelium, the basement membrane, and the astrocyte end-feet.

The blood-brain barrier (BBB) controls what can pass from the blood into the brain tissue. It keeps out many substances, including many drugs and germs, while letting in what neurons need, such as glucose and oxygen. It is a structure built around the brain's capillaries, with three components.

The three components, from the blood outward

Capillary endothelium

The first and most important component. The endothelial cells that line brain capillaries are joined by tight junctions, which seal the gaps between neighboring cells. Brain capillaries also lack the pores and wide clefts found in many capillaries elsewhere in the body. As a result, most substances can only cross by passing through the endothelial cells themselves.

Basement membrane

The second component, a thin layer of extracellular material that surrounds the endothelium.

Astrocyte end-feet

The third component. Astrocytes are star-shaped glial cells. Their processes end in flattened feet that wrap around the outside of the capillary, covering the basement membrane. The end-feet support the endothelial cells and help keep the tight junctions in place.

The barrier is missing in a few small areas around the ventricles, such as parts of the hypothalamus, where the brain needs to sample the blood directly.

Component	What it is	Position from the blood	Its part in the barrier
Capillary endothelium	The cells lining the brain capillaries	First, in contact with the blood	Joined by tight junctions and lacking pores, so substances cannot slip between the cells
Basement membrane	A thin layer of extracellular material around the endothelium	Second	Surrounds and supports the capillary wall
Astrocyte end-feet	The flattened ends of astrocyte processes	Third, the outermost layer	Wrap the outside of the capillary and help maintain the tight junctions

- Memory tip: go from the blood outward: endothelium, then basement membrane, then astrocyte feet. The tight junctions in the endothelium are the actual seal.

- A substance leaving the blood for a neuron crosses the endothelial cells with their tight junctions first, the basement membrane second, and the astrocyte end-feet third. The end-feet are the outermost layer, outside the basement membrane.
- An ordinary capillary has pores in its wall. A brain capillary has none, and that is the difference you draw side by side.
- The end-feet come from astrocytes, not from oligodendrocytes or neurons.

Worked questions

Eight questions in the style of the lecture exam. Read the answer and the reasons each wrong option is wrong. The wrong options are where the learning is.

QUESTION 1 · COMPETENCY 167

Which dural reflection separates the cerebrum from the cerebellum?

A. Falx cerebri

B. Falx cerebelli

C. Tentorium cerebelli

D. Septum pellucidum

Answer: C. The tentorium cerebelli lies in the transverse fissure like a tent over the cerebellum. It roofs the cerebellum and separates it from the occipital lobes of the cerebrum above.

Why the others are wrong

- **A** The falx cerebri hangs in the longitudinal fissure and separates the right and left cerebral hemispheres. It runs front to back, not across.
- **B** The falx cerebelli is the small fold below the tentorium that separates the right and left cerebellar hemispheres. It splits the cerebellum; it does not separate it from the cerebrum.
- **D** The septum pellucidum is not a dural fold. It is the thin sheet of brain tissue between the two lateral ventricles.

QUESTION 2 · COMPETENCY 167

Blood in the superior sagittal sinus flows into which structure next?

A. Confluence of sinuses

B. Sigmoid sinus

C. Internal jugular vein

D. Transverse sinus

Answer: A. The superior sagittal sinus runs front to back along the top edge of the falx cerebri and ends at the confluence of sinuses, at the internal occipital protuberance. From there the blood goes transverse sinus, sigmoid sinus, internal jugular vein.

Why the others are wrong

- **B** The sigmoid sinus is the S-shaped sinus that curves down from the transverse sinus to the jugular foramen. It is two steps later.
- **C** The internal jugular vein is the last step, where the sigmoid sinus leaves the skull through the jugular foramen.
- **D** The transverse sinuses run laterally from the confluence, so the confluence comes first.

QUESTION 3 · COMPETENCY 168

Which space holds cerebrospinal fluid?

A. Epidural space

B. Subdural space

C. Between the periosteal and meningeal dura

D. Subarachnoid space

Answer: D. The subarachnoid space, between the arachnoid and the pia, is a real space that holds CSF, the web-like strands of the arachnoid, and the larger surface blood vessels. A lumbar puncture draws CSF from it.

Why the others are wrong

- **A** The spinal epidural space, between the dura and the bone, holds fat and veins. In the cranium it is only a potential space.
- **B** The subdural space, between the dura and the arachnoid, is a potential space with only a thin film of fluid.
- **C** Where the two dural layers part they form the dural venous sinuses, which carry venous blood. CSF enters them only after the arachnoid granulations have returned it to the blood.

QUESTION 4 · COMPETENCY 168

The denticulate ligaments are made of which meningeal layer, and what do they attach the cord to?

A. Dura mater; the vertebrae

B. Pia mater; the dura mater

C. Arachnoid mater; the pia mater

D. Pia mater; the coccyx

Answer: B. The denticulate ligaments are tooth-shaped flaps of pia along each side of the cord. They reach out through the arachnoid and attach to the dura, which keeps the cord centered in the dural sac.

Why the others are wrong

- **A** The spinal dura is not attached to the vertebrae at all. It is a free tube, the dural sac, inside the fat-filled epidural space.
- **C** The arachnoid sends fine strands to the pia, but those are the web-like strands in the subarachnoid space, not the denticulate ligaments.
- **D** Pia to the coccyx describes the filum terminale, the strand that anchors the lower tip of the cord. The denticulate ligaments run sideways to the dura.

QUESTION 5 · COMPETENCY 169

During a lumbar puncture, the needle passes the ligamentum flavum. What is the next layer it enters?

A. Epidural space

B. Interspinous ligament

C. Dura mater

D. Subarachnoid space

Answer: A. The ligamentum flavum is the last and deepest of the three ligaments, and it joins the laminae. Right after the clinician feels it give, the needle is in the epidural space, the fat-filled and vein-filled space between the bone and the dura. Then come the dura, the arachnoid, and the subarachnoid space.

Why the others are wrong

- **B** The interspinous ligament is crossed before the ligamentum flavum, not after. The order is supraspinous, interspinous, ligamentum flavum.
- **C** The dura is crossed after the epidural space. The needle has to pass the fat and veins before it reaches the dural sac.
- **D** The subarachnoid space is the goal, but the needle still has to cross the epidural space, the dura, and the arachnoid to get there.

QUESTION 6 · COMPETENCY 170

Which opening connects the third ventricle to the fourth ventricle?

A. Interventricular foramen

B. Median aperture

C. Lateral aperture

D. Cerebral aqueduct

Answer: D. The cerebral aqueduct (aqueduct of Sylvius) is the narrow channel through the midbrain. The third ventricle drains down through it into the fourth ventricle.

Why the others are wrong

- **A** The interventricular foramina (of Monro) connect each lateral ventricle to the third ventricle, one on each side.
- **B** The median aperture (foramen of Magendie) is one of the exits from the fourth ventricle into the subarachnoid space, not an entrance to it.
- **C** The two lateral apertures (foramina of Luschka) also lead out of the fourth ventricle into the subarachnoid space.

QUESTION 7 · COMPETENCY 171

Which cells produce cerebrospinal fluid?

A. Astrocyte end-feet

B. Ependymal cells of the choroid plexus

C. Arachnoid mater cells of the arachnoid granulations

D. Endothelial cells of the dural venous sinuses

Answer: B. A choroid plexus is a tuft of capillaries covered by ependymal cells hanging into a ventricle. The ependymal cells make CSF from the blood plasma in those capillaries, most of it in the lateral ventricles.

Why the others are wrong

- **A** Astrocyte end-feet are the outermost component of the blood-brain barrier, wrapped around brain capillaries. They do not make fluid.
- **C** The arachnoid granulations are where CSF is reabsorbed into the dural sinuses, the end of the route, not the start.
- **D** The dural venous sinuses carry venous blood out of the skull. CSF joins that blood only after passing through the arachnoid granulations.

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A molecule leaving a brain capillary for a neuron crosses the three components of the blood–brain barrier in which order?

A. Astrocyte end-feet, basement membrane, endothelium

B. Basement membrane, endothelium, astrocyte end-feet

C. Endothelium, basement membrane, astrocyte end-feet

D. Endothelium, astrocyte end-feet, basement membrane

Answer: C. Counting outward from the blood, the tight-junctioned endothelium is first, the basement membrane that wraps it is second, and the astrocyte end-feet covering the outside are third. The tight junctions in the endothelium are the actual seal.

Why the others are wrong

- **A** This is the order reversed. The astrocyte end-feet are the outermost layer, so they are the last thing crossed, not the first.
- **B** The endothelium lines the capillary and is in direct contact with the blood, so nothing can be crossed before it.
- **D** The basement membrane sits directly on the endothelium, with the astrocyte end-feet outside it, so the end-feet cannot come before the basement membrane.